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1. A meter rule has an accuracy of:

A. ±1.0 cm
B. ±0.1 cm
C. ±0.01 cm
D. ±0.001 cm

Answer: B

Meter rules measure accurately to ±0.1 cm or 1 mm.

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2. The accuracy of a vernier caliper is:

A. ±1 mm
B. ±0.01 cm
C. ±0.1 mm
D. ±0.001 cm

Answer: C

Vernier calipers have an accuracy of 0.1 mm (0.01 cm).

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3. A micrometer screw gauge is best used to measure:

A. Diameter of a wire
B. Length of a book
C. Volume of a liquid
D. Weight of an object

Answer: A

Micrometer screw gauges measure small diameters or thicknesses, e.g., a wire.

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4. The unit of force measured by a spring balance is:

A. Kilograms (kg)
B. Grams (g)
C. Newtons (N)
D. Joules (J)

Answer: C

Spring balances measure force in Newtons (N).

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5. A G-clamp is used to:

A. Measure current
B. Tighten or hold objects in place
C. Measure voltage
D. Determine density

Answer: B

G-clamps secure objects during work like cutting or gluing.

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6. Which instrument is used to measure voltage?

A. Ammeter
B. Spring balance
C. Voltmeter
D. Micrometer screw gauge

Answer: C

A voltmeter measures voltage (potential difference).

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7. To convert 5 cm to millimeters (mm), you:

A. Multiply by 100
B. Divide by 10
C. Multiply by 10
D. Divide by 100

Answer: C

Multiply by 10: .

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8. The mass of an object is measured using:

A. Spring balance
B. Lever balance
C. Ammeter
D. G-clamp

Answer: B

Mass is measured using a lever balance or similar devices.

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9. The unit for density is:

A. kg/m^3
B. N/m^2
C. J/m^2
D. $kg \cdot m/s$

Answer: A

Density = Mass/Volume, and its unit is kg/m³.

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10. Archimedes' principle states that:

A. The force on an object is equal to mass × gravity.
B. A body experiences a force proportional to density.
C. Upthrust equals the weight of the displaced fluid.
D. Pressure depends on volume.

Answer: C

Archimedes’ principle states upthrust = weight of displaced fluid.

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11. A solid weighs 20 N in air and 12 N in water. The upthrust is:

A. 8 N
B. 12 N
C. 20 N
D. 32 N

Answer: A

Upthrust = Weight in air - Weight in water = .

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12. What is measured using a scale pan?

A. Voltage
B. Mass
C. Pressure
D. Force

Answer: B

A scale pan is used to measure mass.

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13. A beam balance operates on:

A. Archimedes' principle
B. Principle of moments
C. Ohm's law
D. Pascal’s law

Answer: B

A beam balance works on the principle of moments.

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14. The density of a liquid is expressed in:

A. kg/m^2
B. kg/m^3
C. N/m
D. J/m^3

Answer: B

Density = Mass/Volume and its unit is kg/m^3.

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15. A micrometer screw gauge measures to an accuracy of:

A. 0.1 mm
B. 0.01 cm
C. 0.01 mm
D. 0.001 cm

Answer: C

Micrometers measure to 0.01 mm accuracy.

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16. The instrument for measuring electric current is:

A. Ammeter
B. Voltmeter
C. Galvanometer
D. Micrometer

Answer: A

An ammeter measures electric current.

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17. The upthrust on an object is equal to:

A. Its weight in air
B. Volume of the fluid
C. Weight of fluid displaced
D. Its mass

Answer: C

Archimedes' principle: Upthrust = Weight of fluid displaced.

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18. 1 mm equals how many meters?

A. 0.001 m
B. 0.01 m
C. 1.0 m
D. 0.1 m

Answer: A

.

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19. The weight of an object in water is reduced due to:

A. Gravity
B. Buoyant force
C. Pressure
D. Mass

Answer: B

Buoyant force (upthrust) reduces the weight of an object in water.

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20. A spring balance measures:

A. Volume
B. Density
C. Force
D. Mass

Answer: C

A spring balance measures force in Newtons (N).

Experiment 1: Measurement

1. The smallest division on a standard meter rule is:
A. 1.0 cm
B. 0.1 cm

C. 0.01 cm
D. 0.001 cm

Answer: B

The smallest division on a meter rule is 0.1 cm or 1 mm.

2. The accuracy of a vernier caliper is:
A. 0.1 mm
B. 0.01 mm
C. 1.0 mm
D. 0.001 mm

Answer: A

A vernier caliper can measure with an accuracy of 0.1 mm.

3. A micrometer screw gauge is best used to measure:
A. Thickness of a book
B. Diameter of a thin wire
C. Height of a cylinder
D. Volume of water

Answer: B

Micrometers are precise tools for measuring small thicknesses like wires.

4. The least count of a micrometer screw gauge is:
A. 0.01 cm
B. 0.001 cm
C. 0.01 mm
D. 0.1 mm

Answer: C

The least count of a micrometer is typically 0.01 mm.

Experiment 2: Archimedes' Principle

5. Archimedes' Principle states that the upthrust on a body in a fluid is equal to:
A. The weight of the object
B. The weight of the fluid displaced
C. The weight of the object in water
D. The volume of the object

Answer: B

Archimedes' principle states upthrust = weight of the displaced fluid.

6. A body weighs 10 N in air and 8 N when submerged in water. What is the upthrust?
A. 18 N
B. 8 N
C. 10 N
D. 2 N

Answer: D

Upthrust = Weight in air - Weight in water = .

7. The upthrust on a body in water depends on:
A. Mass of the object
B. Density of the object
C. Density of the fluid and volume displaced
D. Weight of the object

Answer: C

Upthrust depends on the density of the fluid and the volume of fluid displaced.

8. A block floats in water when its weight equals:
A. Its volume
B. The weight of displaced water
C. The density of water
D. The buoyant force

Answer: B

Floating occurs when the weight equals the weight of displaced water.

Experiment 3: Determination of Density of a Solid

9. The formula for density is:
A. Mass × Volume
B. Volume / Mass
C. Mass / Volume
D. Force × Volume

Answer: C

Density = .

10. The SI unit of density is:
A. kg/m³
B. g/cm³
C. N/m²
D. kg·m/s²

Answer: A

The standard SI unit of density is kg/m³.

11. To find the volume of an irregular solid, you would use:
A. A ruler
B. A vernier caliper
C. The displacement method
D. A spring balance

Answer: C

The displacement method (water in a measuring cylinder) finds volume.

12. If an object has a mass of 120 g and volume of 40 cm³, its density is:
A. 3 g/cm³
B. 4 g/cm³
C. 6 g/cm³
D. 0.33 g/cm³

Answer: A

Density = Mass/Volume = .

Experiment 6: The Simple Pendulum

13. The time period of a simple pendulum depends on:
A. Amplitude only
B. Mass of the bob
C. Length of the string
D. Density of the bob

Answer: C

Time period depends on the length of the string.

14. The formula for the time period of a simple pendulum is:
A.
B.
C.
D.

Answer: A

, where is length and is acceleration due to gravity.

15. What happens to the time period if the length of a pendulum increases?
A. It decreases
B. It remains the same
C. It increases
D. It becomes zero

Answer: C

The time period increases as the length increases.

16. Which of the following does not affect the time period of a pendulum?
A. Length of the string
B. Amplitude for small angles
C. Gravity
D. Mass of the bob

Answer: D

The mass of the bob does not affect the time period.

Experiment 13: Hooke’s Law

17. Hooke’s Law states that:
A. Extension is directly proportional to mass
B. Extension is inversely proportional to force
C. Force is directly proportional to extension
D. Force is inversely proportional to extension

Answer: C

Hooke’s Law: Force (extension).

18. The formula representing Hooke's Law is:
A. F=Kx
B. F=k/x
C.
D.

Answer: A

, where is the spring constant.

19. A spring stretches by 4 cm when a force of 8 N is applied. The spring constant is:
A. 2 N/cm
B. 0.5 N/cm
C. 4 N/cm
D. 8 N/cm

Answer: A

20. The spring constant is measured in:
A. N/m
B. kg/m³
C. N/s
D. m/N

Answer: A

Spring constant has the unit N/m.